

AI FINANCIAL ADVICE: SUPPLY, DEMAND, AND LIFE CYCLE IMPLICATIONS

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NBER SI Household Finance

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→ Randomize prompt features to isolate supply from demand
Example: Gender gap in investment advice $\approx$ 2/3 demand, 1/3 supply

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State Variables

EXOGENOUS STATES

- Income process
- Employment transitions
- Asset returns
- Mortality risk

SIPP + SSA + CRSP

ENDOGENOUS STATES

- Wealth
- Asset allocation

State Variables



Policy Functions

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- Income process
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SIPP + SSA + CRSP

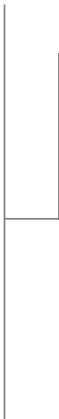
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BENCHMARK I

Life Cycle Model

Choukhmane-de Silva 2026



State Variables

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Simulated Choices

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Choukhmane-de Silva 2026

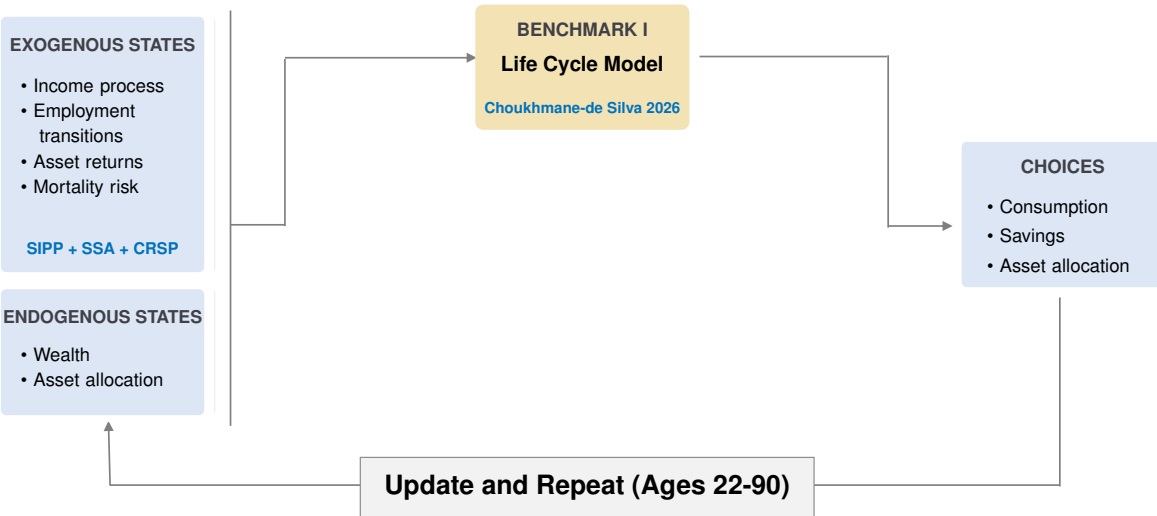
CHOICES

- Consumption
- Savings
- Asset allocation

State Variables

Policy Functions

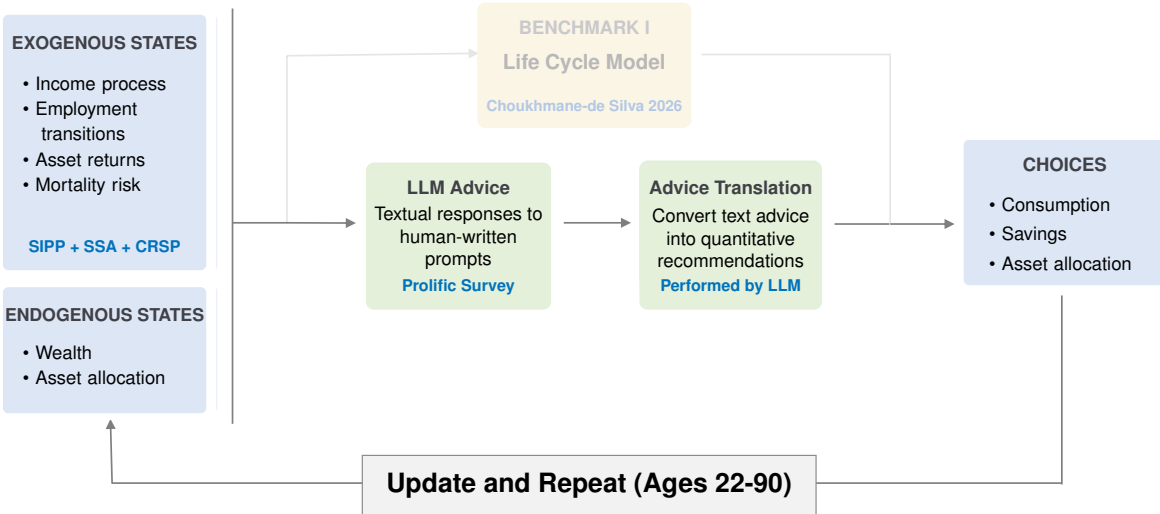
Simulated Choices



State Variables

Policy Functions

Simulated Choices



STATES

PROMPT

ADVICE

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SHOCKS

Simulated Individual

Age: 26

Employment status: Employed

Pre-tax income: \$68,810

Asset allocation:

- Fixed income: \$17,938
- Diversified stocks: \$6,442
- Individual stocks: \$0
- Alternative assets: \$0

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Prolific Prompt (Random by Age-Employment-Income)

Part 1: Financial Situation

I make around 75000 a year before taxes and pay 1700 a month in rent. That is my primary income and I make some money completing paid research surveys. I have around 80000 saved. I am 25.

Part 2: Spending Advice

I am looking for ways to go out more and enjoy my life without sacrificing my savings. I have a good grip on money and actually struggle to justify spending.

Part 3: Investing Advice

How should I put away money now that interest rates are going down. Safe bets like my money market and CDs are no longer yield much money.

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Part 1: Financial Situation

I make around {income} a year before taxes and pay 1700 a month in rent. That is my primary income and I make some money completing paid research surveys. I have around {wealth} saved. I am {age}.

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Prolific Prompt (Random by Age-Employment-Income)

Part 1: Financial Situation

I make around 68810 a year before taxes and pay 1700 a month in rent. That is my primary income and I make some money completing paid research surveys. I have around 24380 saved. I am 26.

Part 2: Spending Advice

I am looking for ways to go out more and enjoy my life without sacrificing my savings. I have a good grip on money and actually struggle to justify spending.

Part 3: Investing Advice

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Prompt Sent to LLM (via API Call with Clear Context)

"I make around 68810 a year before taxes and pay 1700 a month in rent. That is my primary income and I make some money completing paid research surveys. I have around 24380 saved. I am 26.

I am looking for ways to go out more and enjoy my life without sacrificing my savings. I have a good grip on money and actually struggle to justify spending.

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LLM Response

Given your income/savings, you can safely “permission” yourself to spend more without derailing goals by separating **spending from saving**.

- 1) **Set a floor, then automate:** Keep an emergency fund of **3-6 months** of essential expenses in a **HYSA/money market** (don’t chase yield; it’s for safety). Automate investing each payday (e.g., **15-25% of gross**, adjust to comfort). What’s left is guilt-free.
- 2) **Create a “fun” system:** Open a separate checking/savings called “FUN”. Auto-transfer a fixed amount (start **\$300-\$600/month**). If it’s there, you must use it or roll it to a bigger experience.
- 3) **Where to put money as rates fall:** For long-term (5+ years): **low-cost broad index funds/ETFs** (e.g., total US + total international) in 401(k)/Roth IRA first. For medium-term goals (1-5 years): HYSA + Treasuries/T-bill ladder (often tax-advantaged vs CDs), or a short-term bond fund if you can tolerate modest fluctuation.
- 4) **Psychology:** Treat spending from the FUN account as a goal, not a failure-your system already protected savings.

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LLM Advice (Translated with LLM)

Consumption: **\$44,378**
Total savings: **\$17,845**
Asset allocation:
- Fixed income: **+\$5,400**
- Diversified stocks: **+\$12,445**
- Individual stocks: \$0
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Labor Market Transitions

- Age and income specific job-to-job and unemployment transitions
- Persistent and transitory income shocks

Asset Returns

- Fixed income: constant risk-free rate
- Diversified stocks: CRSP Value-Weighted Index
- Individual stocks: CAPM + Bessembinder (2018)
- Alternative assets: individual stock return process with independent realizations

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- Age and income specific job-to-job and unemployment transitions
- Persistent and transitory income shocks

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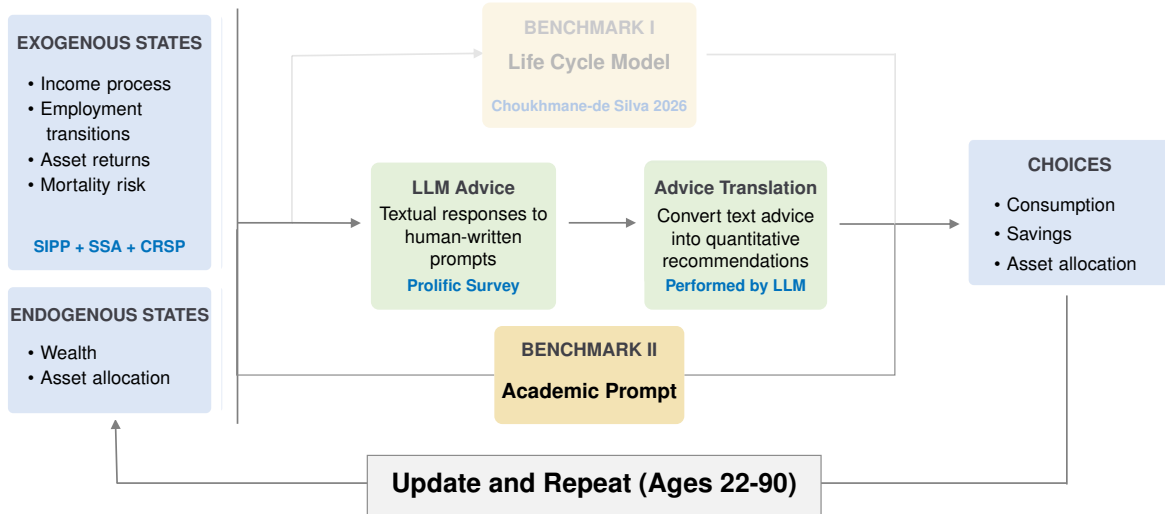
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Academic Prompt (Directly Outputs Numeric Choices)

You are a U.S. financial advisor acting in the best interest of your clients. You are academically trained in household finance, modern portfolio theory, and life cycle planning. **Your objective is to produce high-quality personalized financial advice tailored to my circumstances** under these baseline assumptions:

- Normal life expectancy, living expenditures, retirement age, employment risk, and income risk
- Current U.S. tax law and Social Security rules stay unchanged
- Risk-free savings earn 2% real return annually
- Real stock returns match the 60 year U.S. total stock market historical average
- I am single with no dependents and have no bequest motive. I do not care about wealth after death.

Given my ****age, employment status, annual post-tax salary, average annual post-tax income (since age 22), taxable account balances, and total net wealth****, give the best financial advice for allocating my resources across annual spending and the following four taxable accounts:

[...]

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 - ② *“In your own words, write a prompt asking an AI tool for advice on **how much to spend over the coming year**”*
 - ③ *“In your own words, write a prompt asking an AI tool for advice on **how to invest your savings** between stocks and safer options”*

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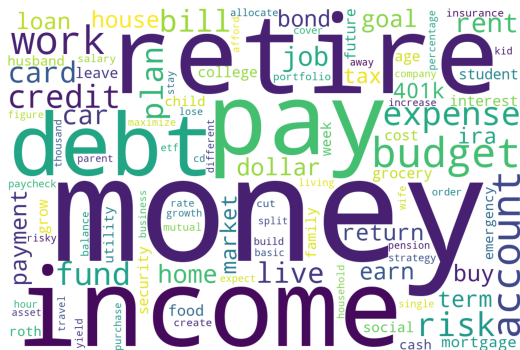
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- Collect demographics, financial literacy, risk/time preferences, saving motives
- Aim for authentic prompts: 952 pass quality check (AI detection + manual specificity)

*“Please write naturally, as you would if you were really asking advice for yourself. There are no right or wrong answers.
Please do not consult outside sources or use AI tools to complete this task”*

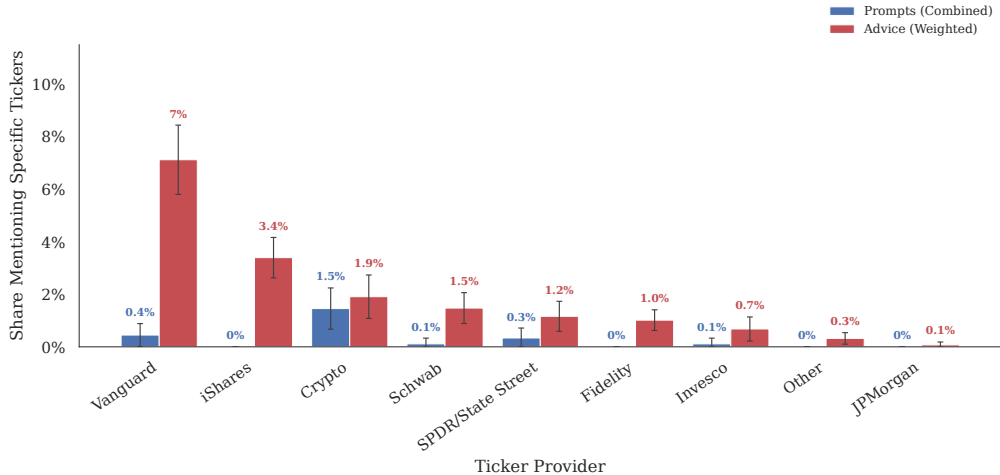
WHAT DO INDIVIDUALS WRITE ABOUT?

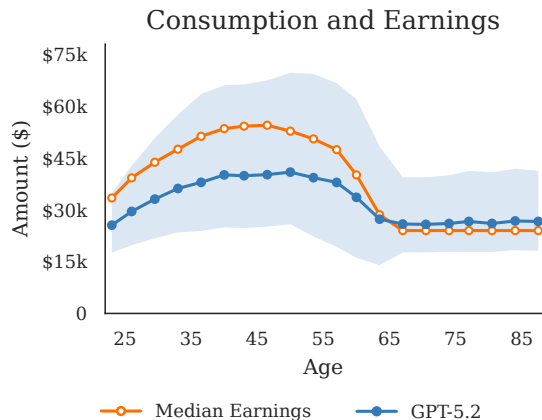
Individuals' Prompts



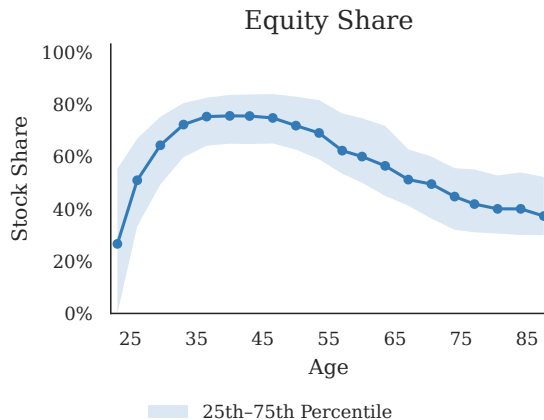
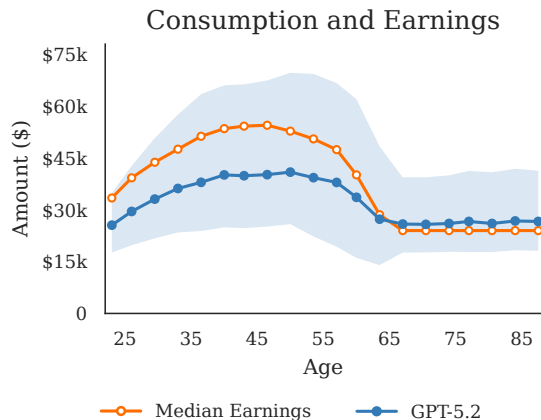
Word Size $\propto$ Pr(word)

WHICH INVESTMENT PRODUCTS ARE DISCUSSED?

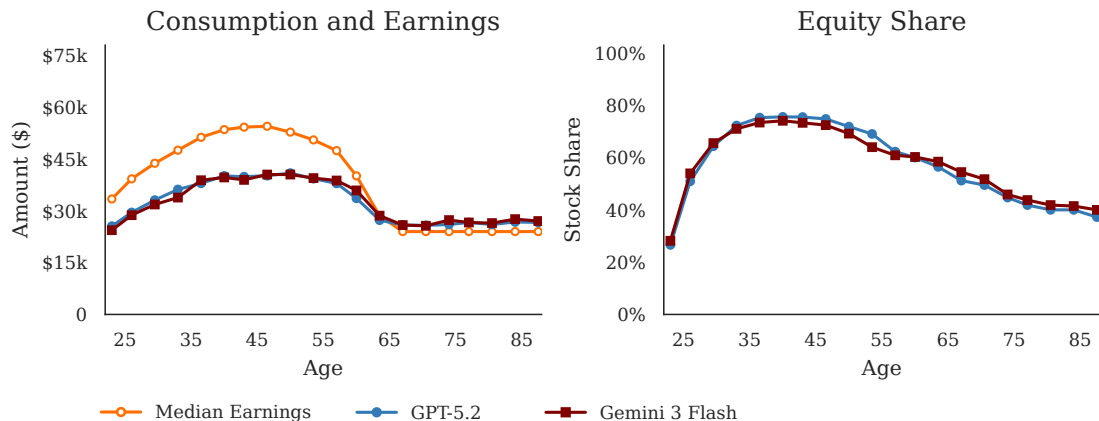




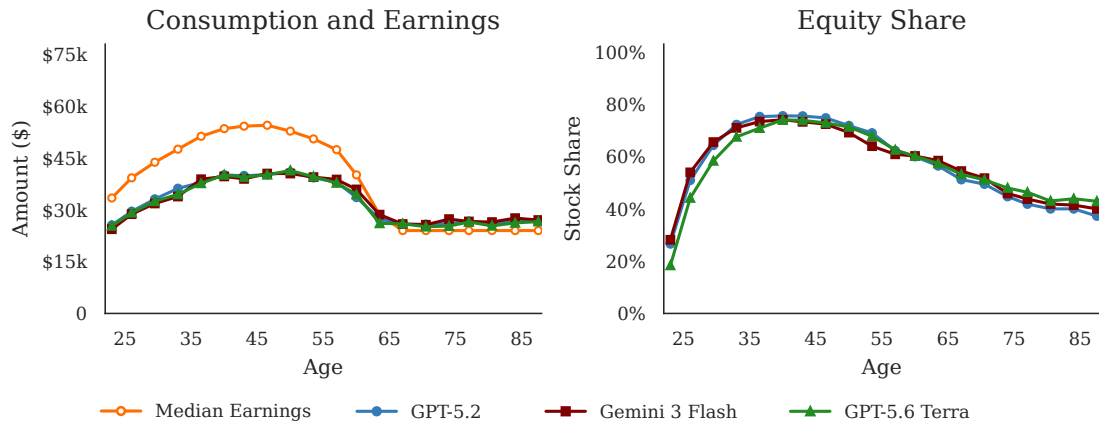
QUANTITATIVE ADVICE: BASELINE SIMULATION



QUANTITATIVE ADVICE: OTHER MODELS



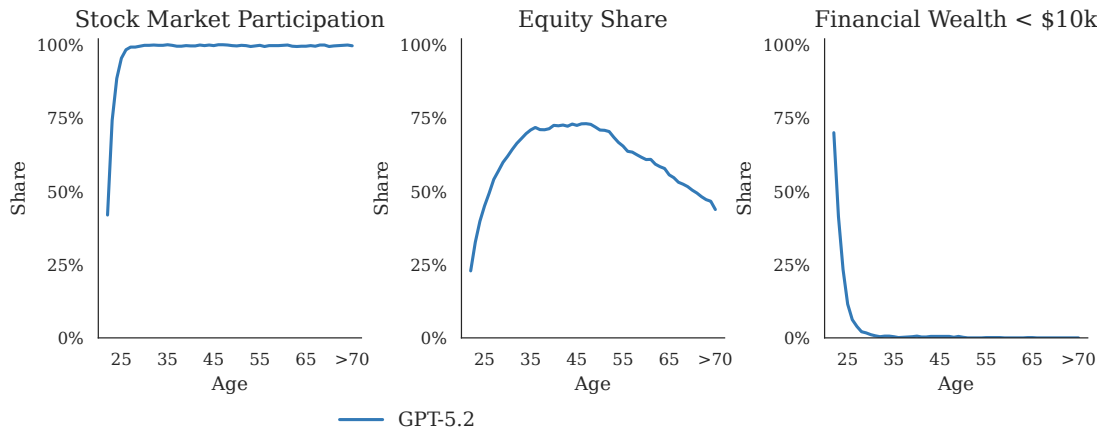
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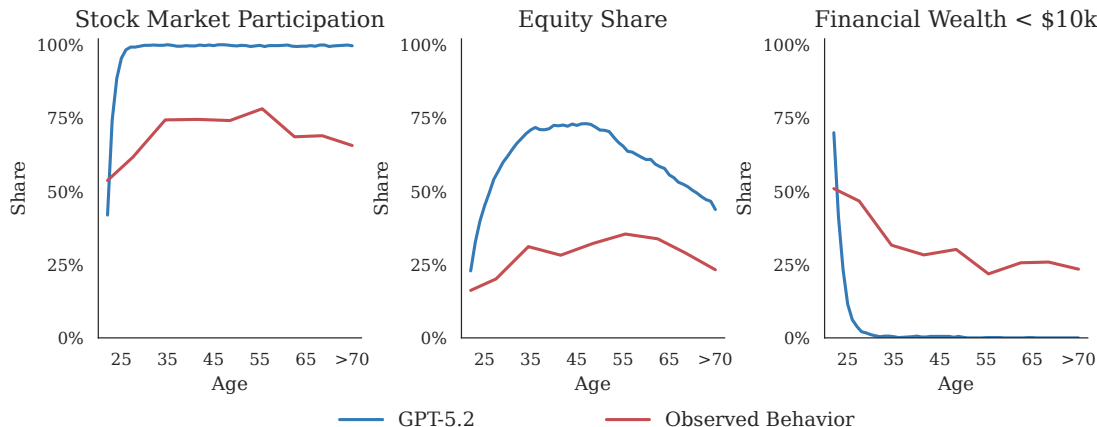
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Fact 1: Advice moves most individuals towards life cycle theory

OBSERVED BEHAVIOR VERSUS LLM ADVICE



OBSERVED BEHAVIOR VERSUS LLM ADVICE



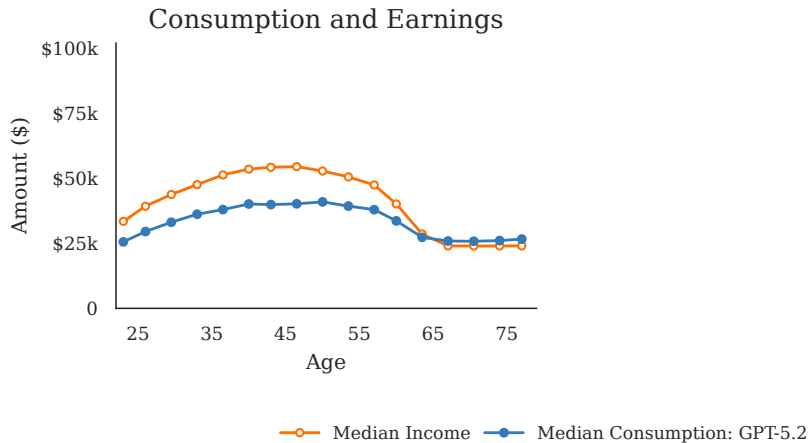
Higher stock market participation, age-declining equity shares after 45, and larger precautionary buffers

THREE FACTS ABOUT LLM FINANCIAL ADVICE

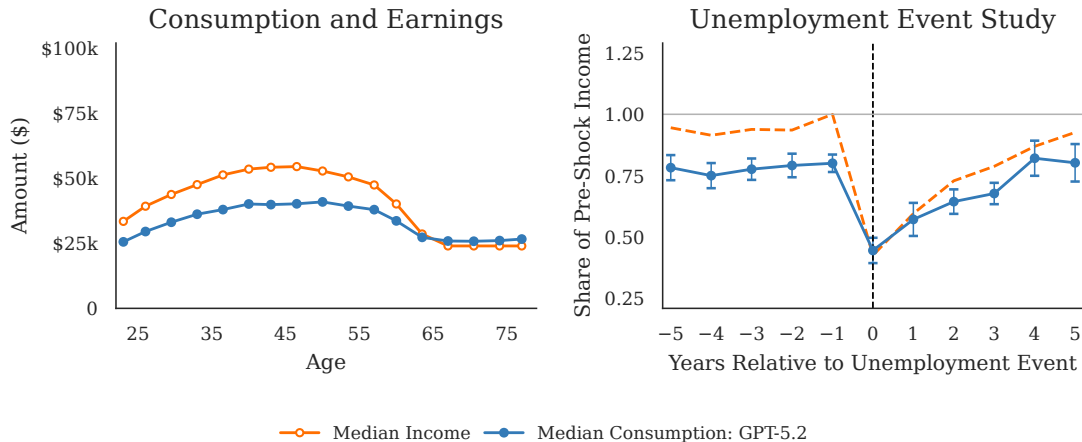
Fact 1: Advice moves most individuals towards life cycle theory

Fact 2: Advice departs from life cycle theory in systematic ways

LLM ADVICE INSUFFICIENTLY SMOOTHS CONSUMPTION

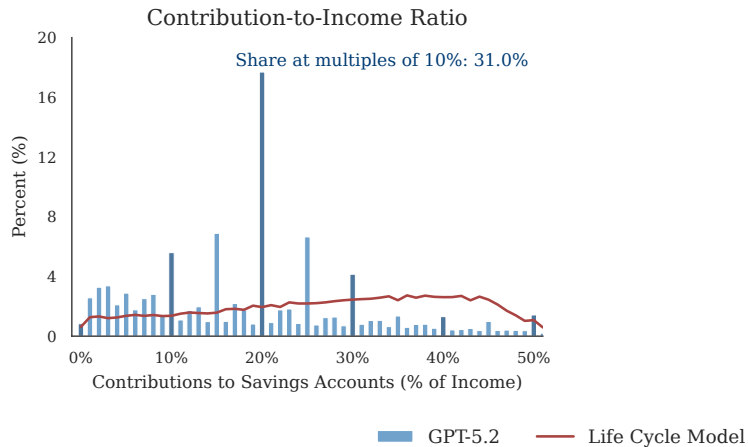


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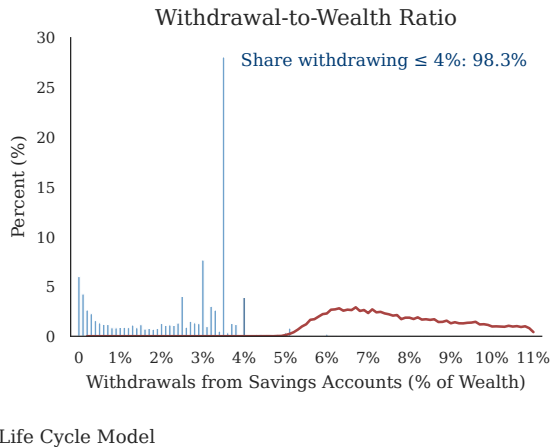
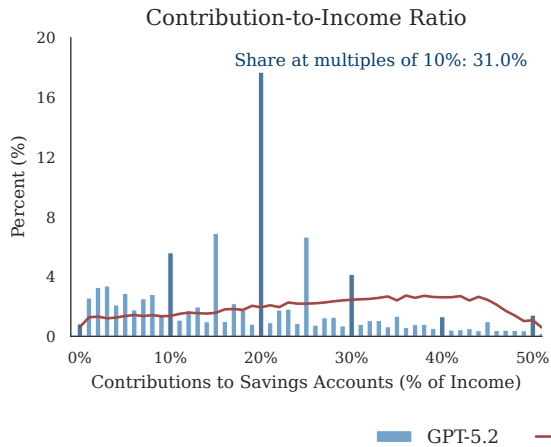
Consumption drops at unemployment despite large buffer stocks (Gruber 1997, Ganong-Noel 2019)

LLM ADVICE RELIES ON HEURISTICS



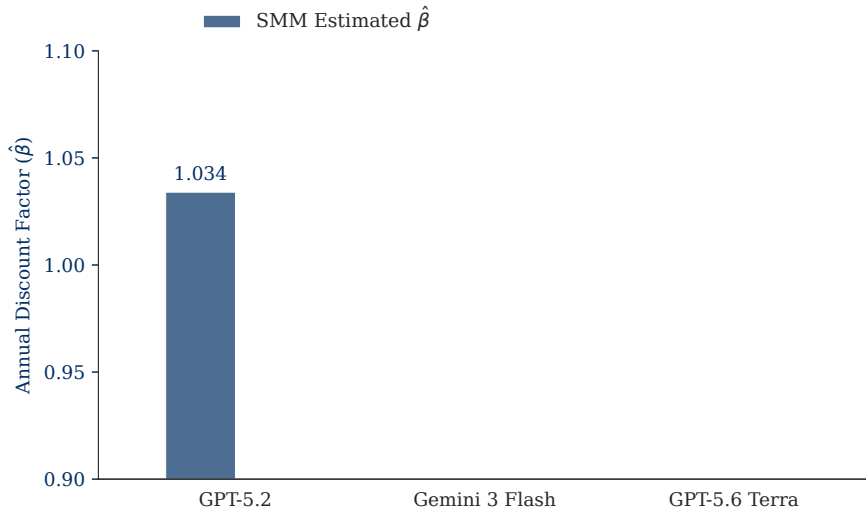
Savings rates bunch at round numbers...

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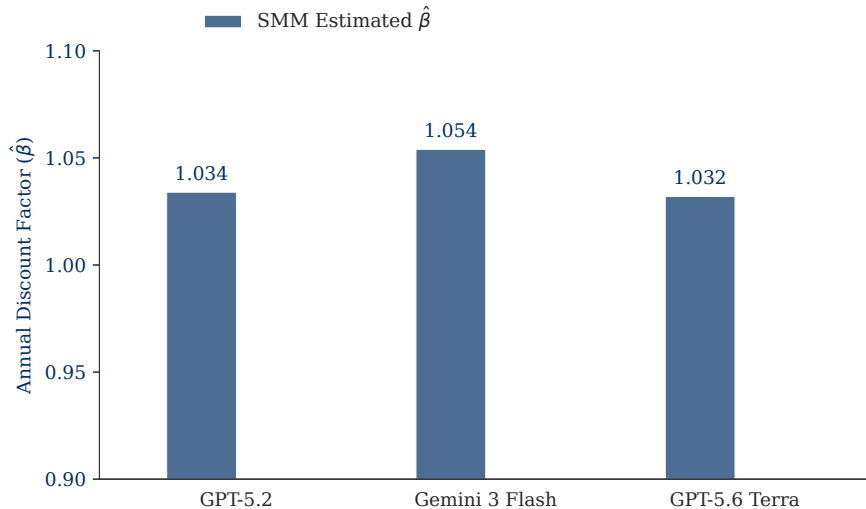


... and withdrawals respect the popular 4% withdrawal rule (Cooley et al. 1998)

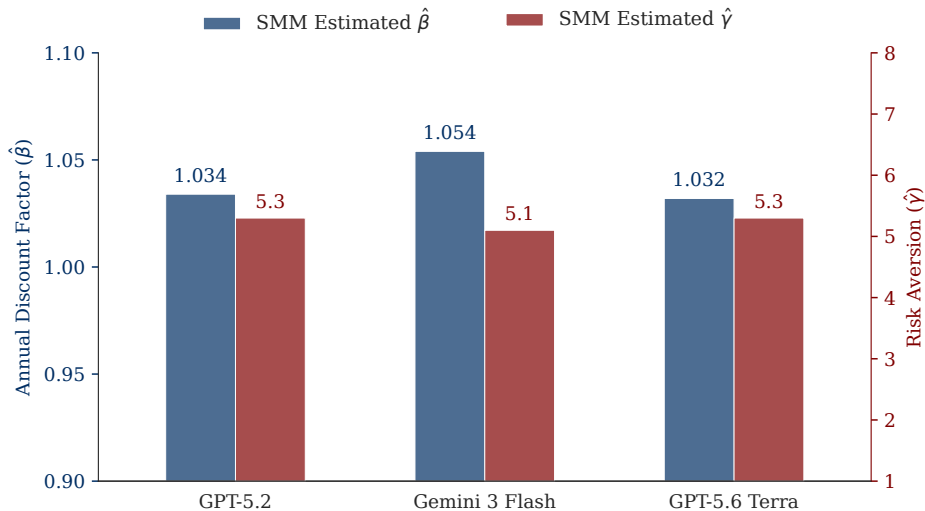
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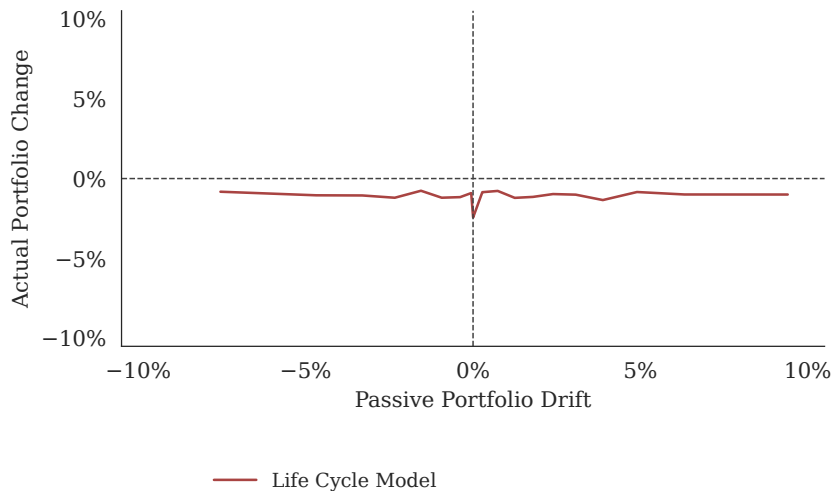
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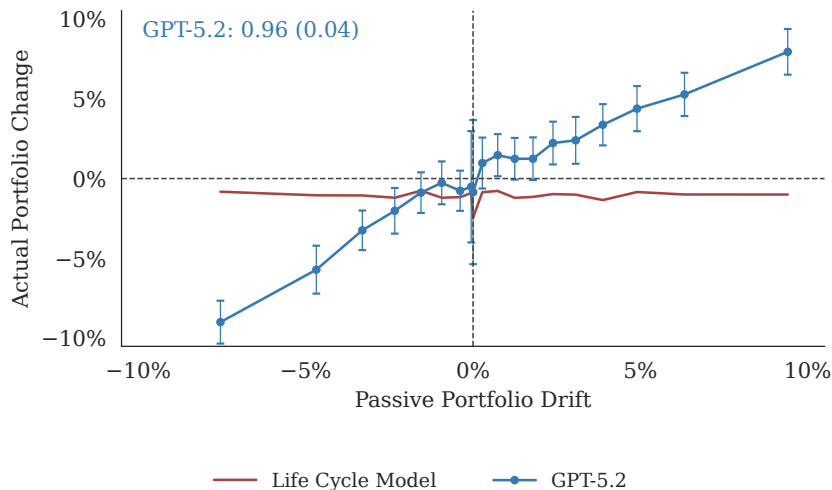
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LLM PORTFOLIOS EXHIBIT SUBSTANTIAL INERTIA



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Equity shares drift passively with realized returns

(Calvet et al. 2009)

ACADEMIC PROMPT REDUCES SOME DEPARTURES FROM THEORY

- ① **Fewer heuristics:** No bunching at round numbers + more withdrawals
- ② **Better consumption-smoothing** over the life cycle and during unemployment
- ③ **More realistic patience:** SMM estimates become $\hat{\beta} = 0.99$ and $\hat{\gamma} = 4.7$
- ④ **No change in portfolio inertia:** Actual–passive portfolio slope unchanged

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⇒ LLM advice depends on characteristics of **prompts** (demand)

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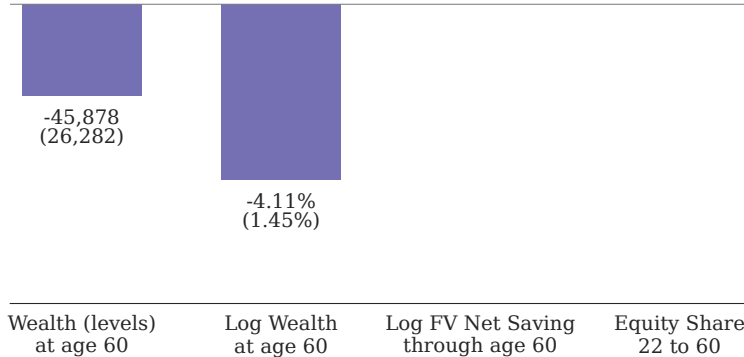
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Fact 2: Advice departs from life cycle theory in systematic ways

Fact 3: Advice varies systematically across groups

LOW-FINANCIAL-LITERACY PROMPTS ACCUMULATE LESS WEALTH

Financial Literacy: Low vs. High

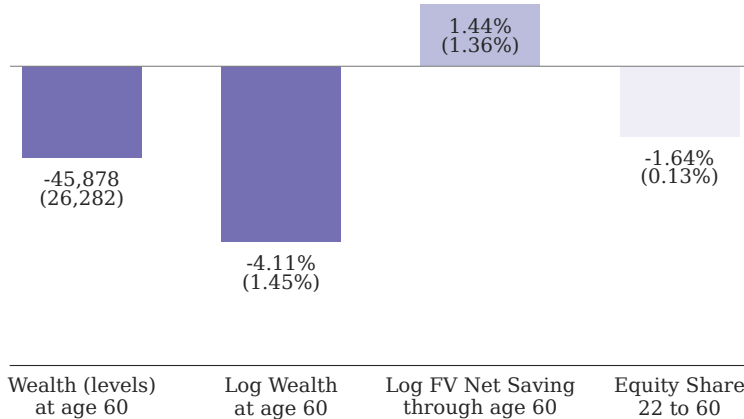


Note: Identical realization of shocks across simulations

(Lusardi and Mitchell 2023)

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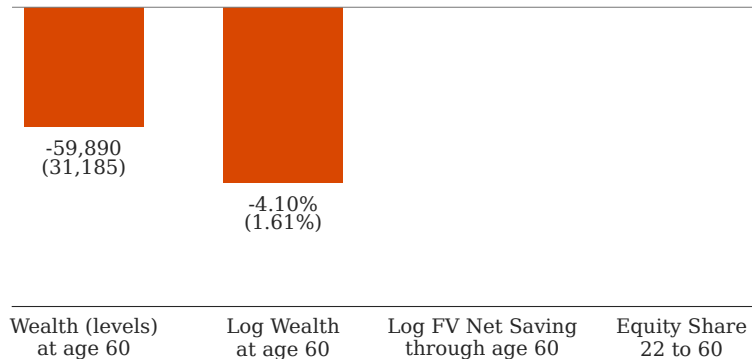


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PROMPTS FROM WOMEN ACCUMULATE LESS WEALTH

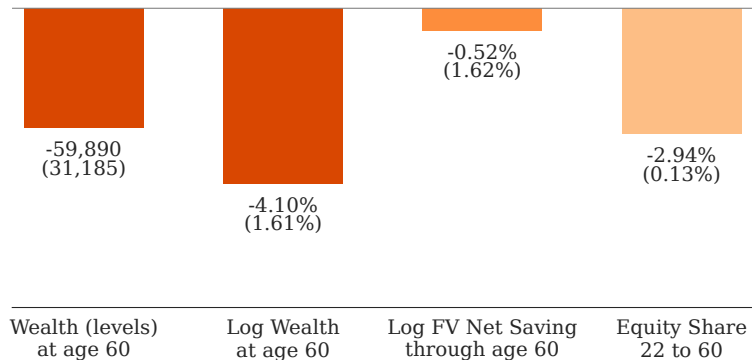
Gender: Female vs. Male



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- 1 Methodology for Simulating LLM Financial Advice
- 2 Describing Prompts and Baseline LLM Financial Advice
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- 4 Heterogeneity in LLM Financial Advice: Supply vs. Demand**
- 5 Conclusion

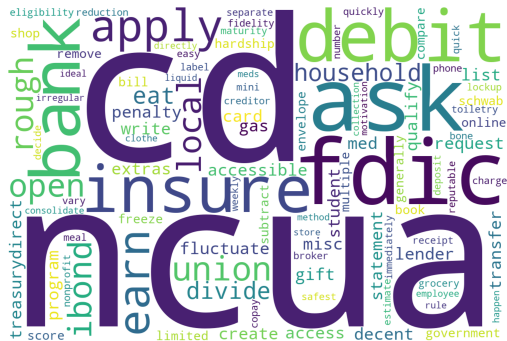
Are these Differences Driven by
Demand (Questions Asked)
or **Supply** (Model Response)?

DIFFERENCES IN LLM RESPONSES BETWEEN MEN AND WOMEN

Men



Women



Word Size $\propto \Pr(\text{word} \mid \text{group}) - \Pr(\text{word} \mid \text{other group})$

Experiment: Consider prompts that do not identify gender (83%) and randomly insert:

“I am a man” or *“I am a woman”*

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Specification:

$$Y_{i,t} = \beta_D \text{FemaleAuthor}_i + \beta_S \text{FemaleLabel}_{i,t} + \delta_{i,t}^{\text{bucket}} + \varepsilon_{i,t}$$

- $\text{FemaleAuthor}_i = 1$: prompt *actually* written by a woman
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	Male Label	Female Label
Male Author	0 (reference)	
Female Author		

SEPARATING SUPPLY AND DEMAND: METHODOLOGY

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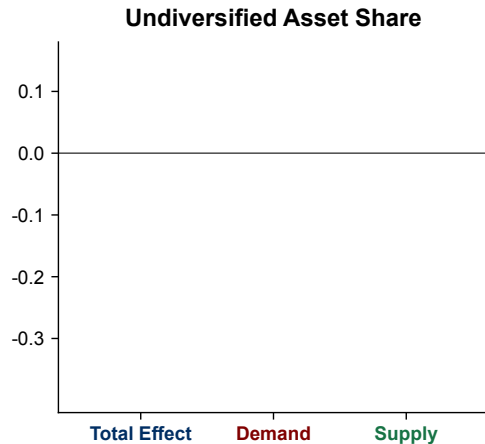
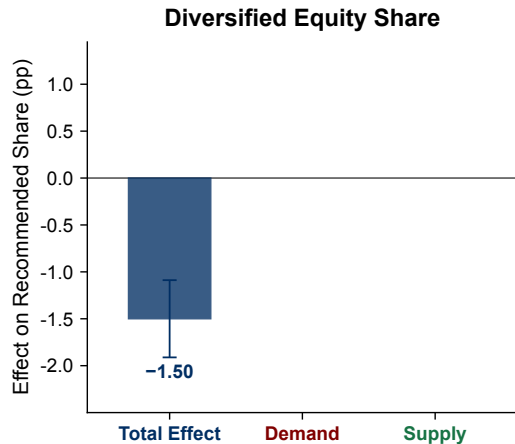
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	Male Label	Female Label
Male Author	0 (reference)	β_S (supply)
Female Author	β_D (demand)	$\beta_D + \beta_S$ (total)

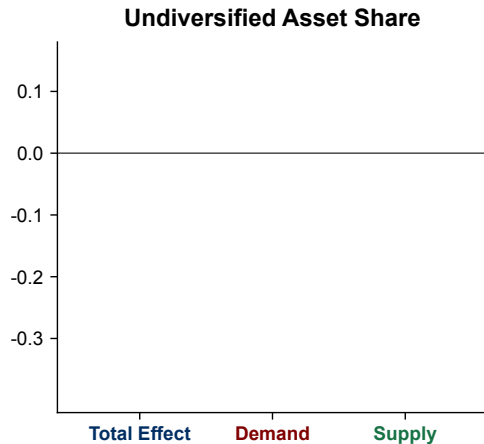
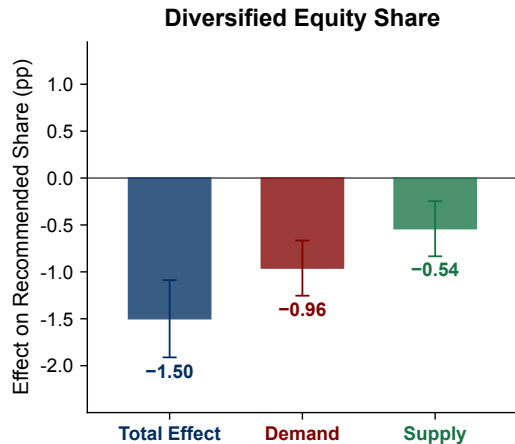
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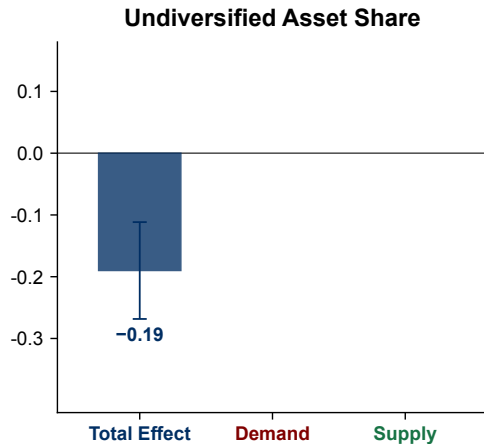
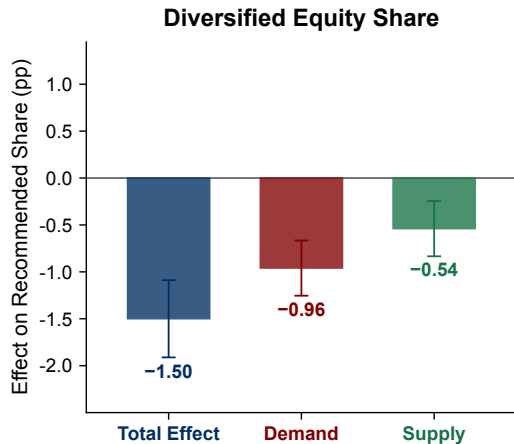
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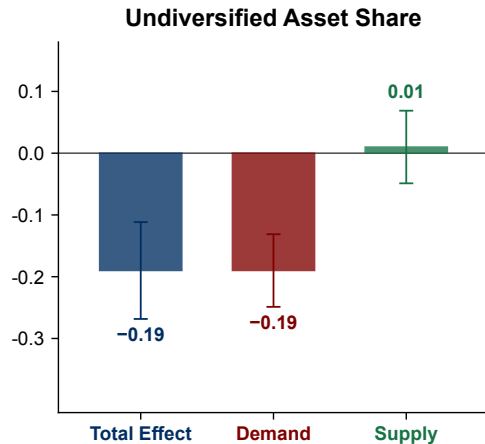
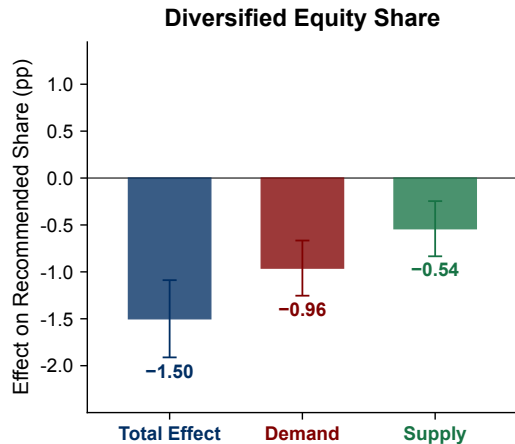
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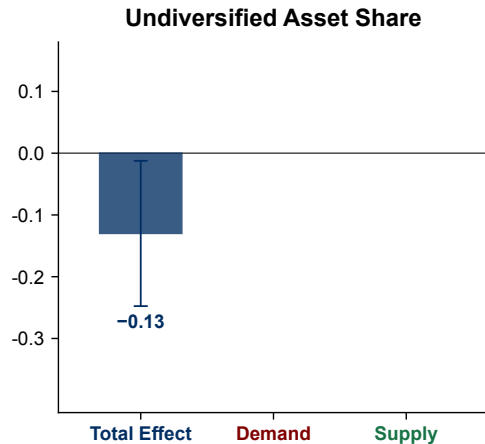
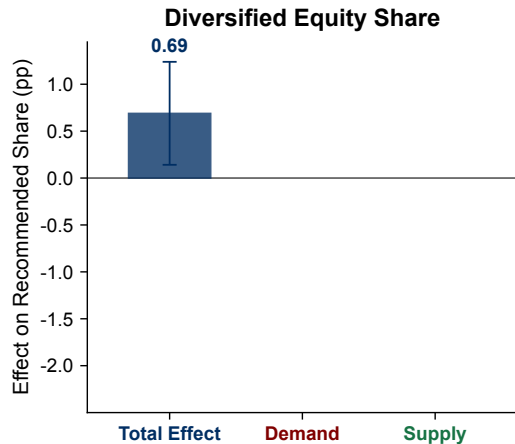
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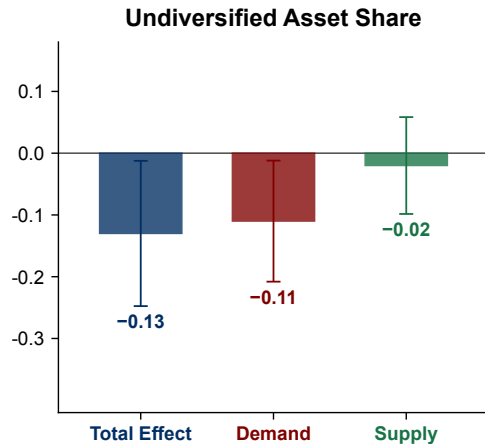
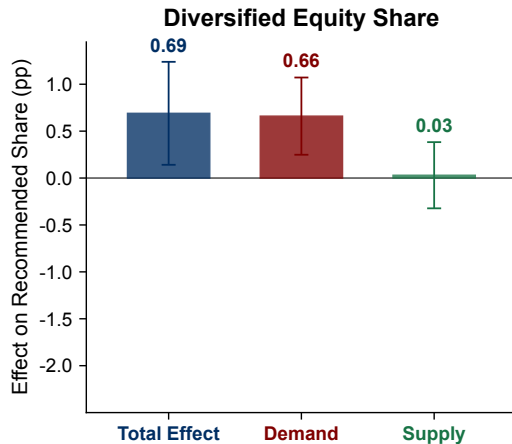
SEPARATING SUPPLY AND DEMAND: BLACK VERSUS WHITE

$$Y_{i,t} = \beta_D \text{BlackAuthor}_i + \beta_S \text{BlackLabel}_{i,t} + \delta_{i,t}^{\text{bucket}} + \varepsilon_{i,t}$$



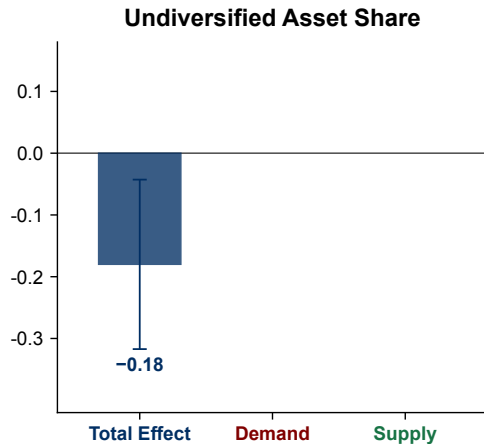
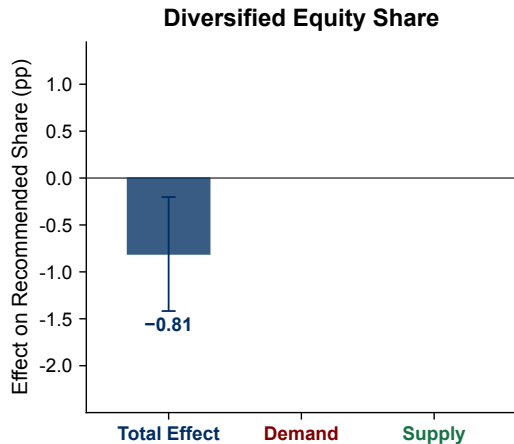
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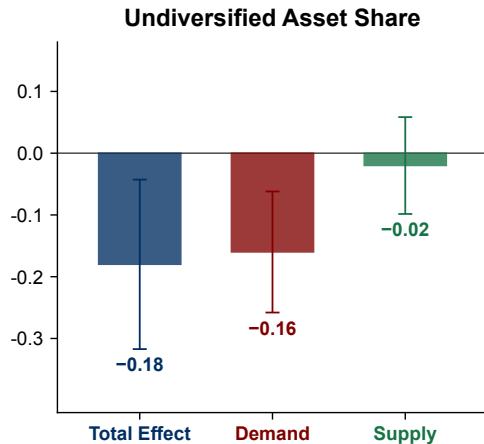
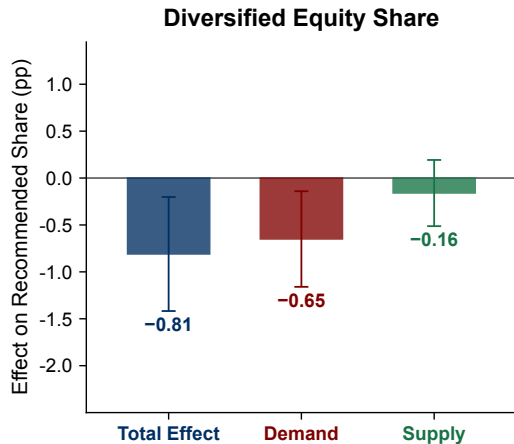
SEPARATING SUPPLY AND DEMAND: ASIAN VERSUS WHITE

$$Y_{i,t} = \beta_D \text{AsianAuthor}_i + \beta_S \text{AsianLabel}_{i,t} + \delta_{i,t}^{\text{bucket}} + \varepsilon_{i,t}$$



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Separating **supply** from **demand** is **normatively important**

- **Supply**-driven differences may or may not be desirable
 - **Implicit inference**: demographics correlate with advice-relevant characteristics
 - Or **algorithmic bias**: reproduces training-data stereotypes
- **Supply** shaped by design: no race effect likely reflects **training safeguards**
- **Demand**-driven differences reflect what people actually write about
 - Good for **personalization**, but may amplify existing biases

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Fact 1: Advice moves most individuals towards life cycle theory

Fact 2: Advice departs from life cycle theory in systematic ways

Fact 3: Advice varies systematically across groups

- **Takeaway #1:** LLM financial advice can improve decision-making
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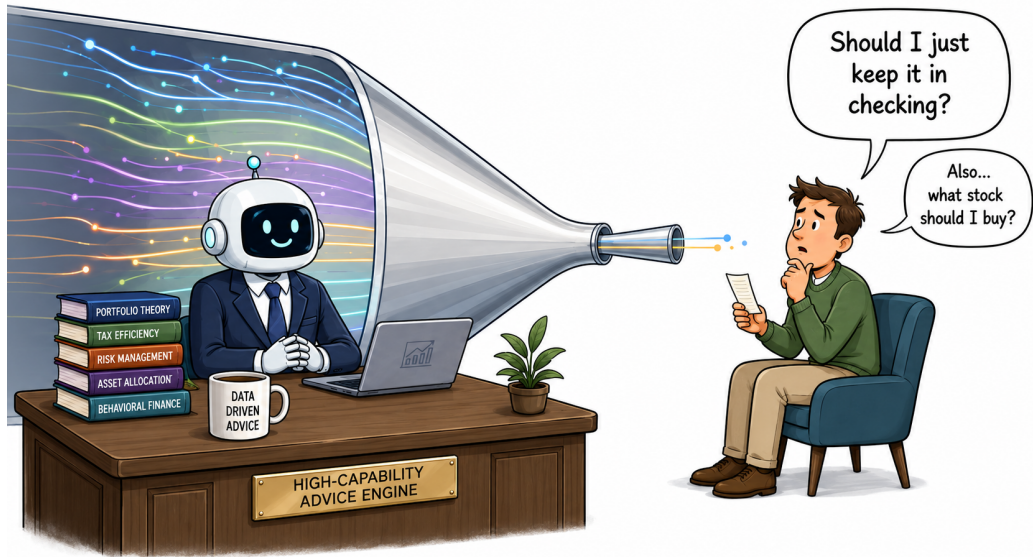
How will these results change as models evolve?

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How will these results change as models evolve?

This paper:

- ① **Scalable methodology** to track model and user evolution
- ② **Diagnostic tests grounded in life cycle theory** that can guide model design
- ③ **Demand-side** will likely evolve more slowly, remaining a binding constraint



Source: ChatGPT-5.6